

Операционные системы

Командная оболочка Midnight Commander

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. Процесс выполнения лабораторной работы

Работа с Midnight Commander

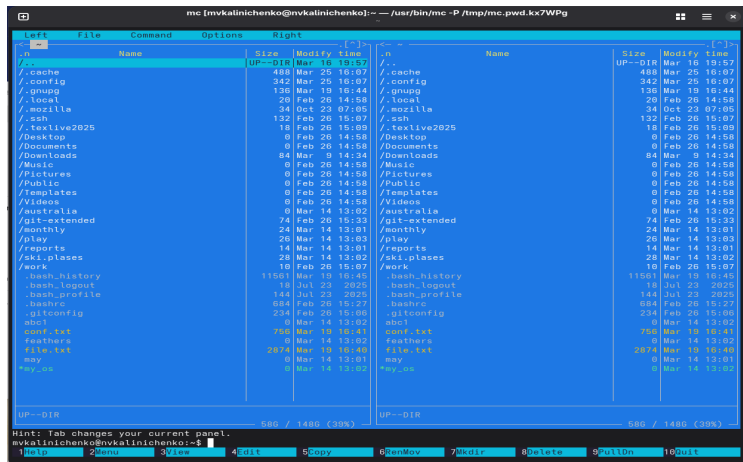


Рисунок 1: Запуск mc

Работа с Midnight Commander

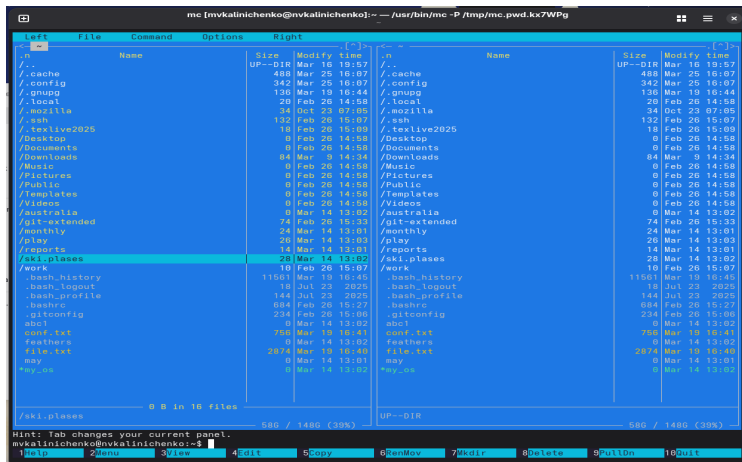


Рисунок 2: Выделение

Работа с Midnight Commander

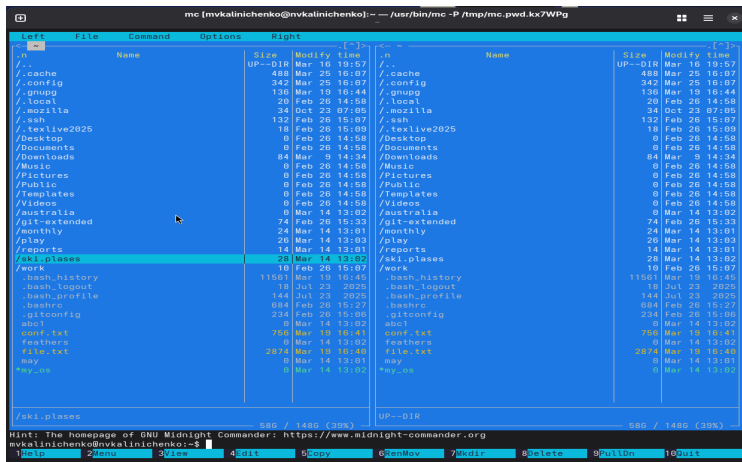


Рисунок 3: Отмена

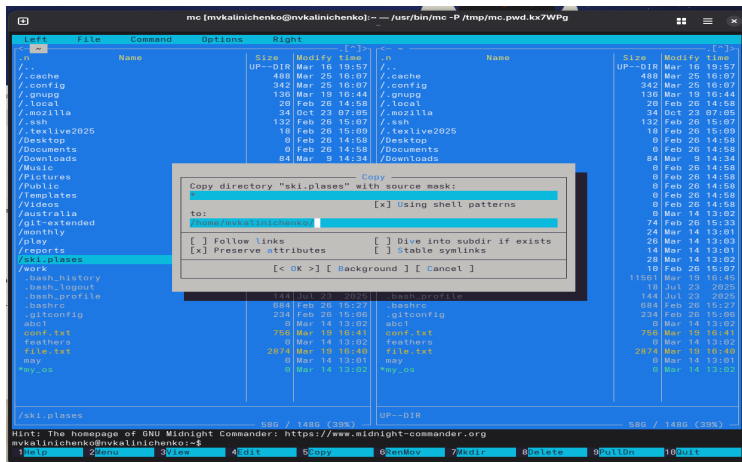


Рисунок 4: Копирование

Работа с Midnight Commander

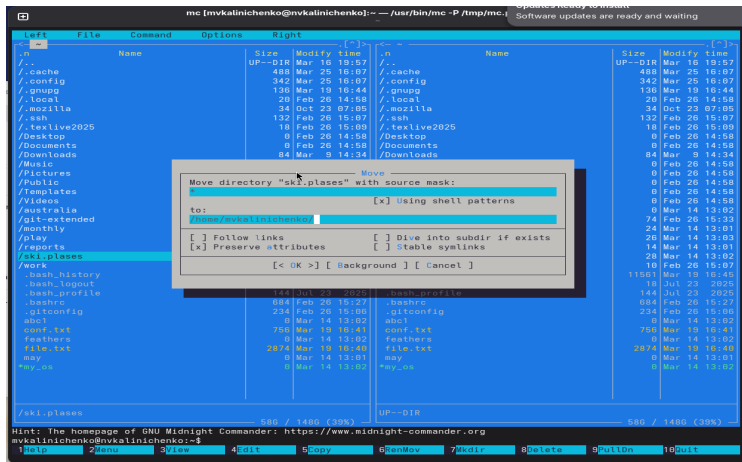


Рисунок 5: Перемещение

Работа с Midnight Commander

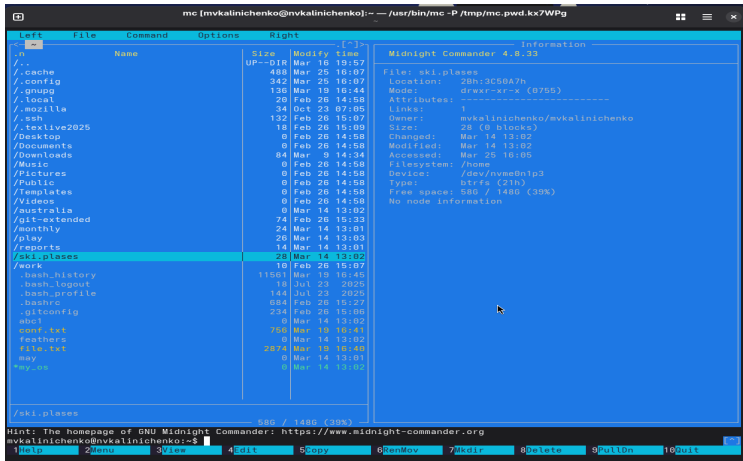


Рисунок 6: Информация

Работа с Midnight Commander

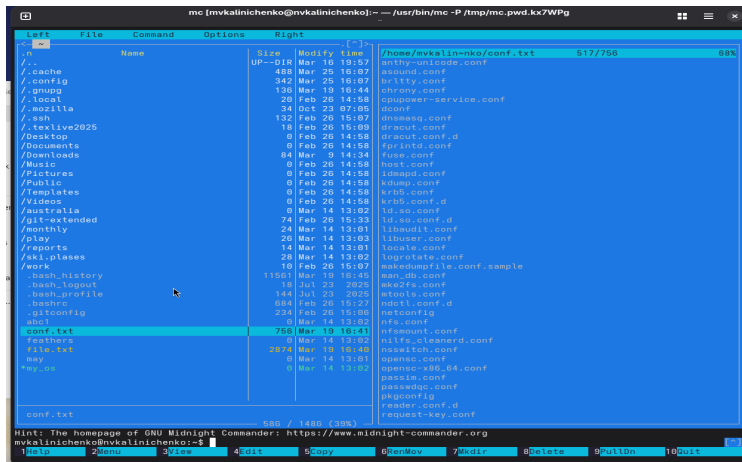


Рисунок 7: Быстрый просмотр

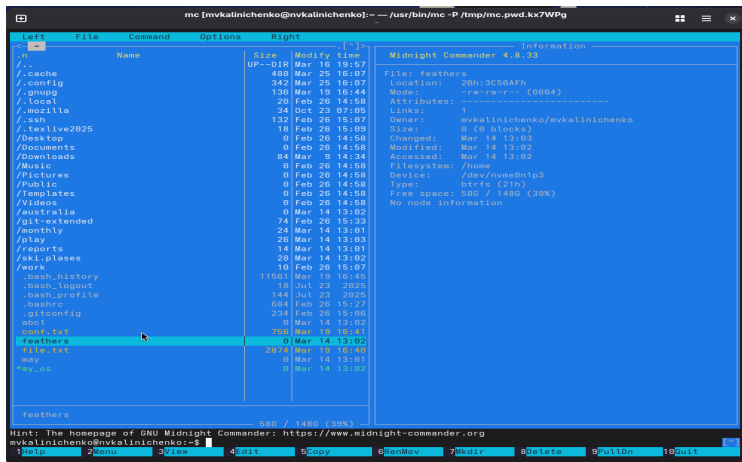


Рисунок 8: Информация

Работа с Midnight Commander

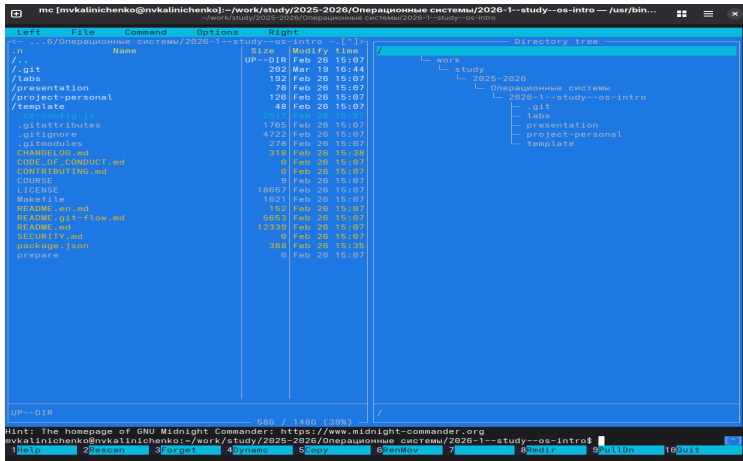
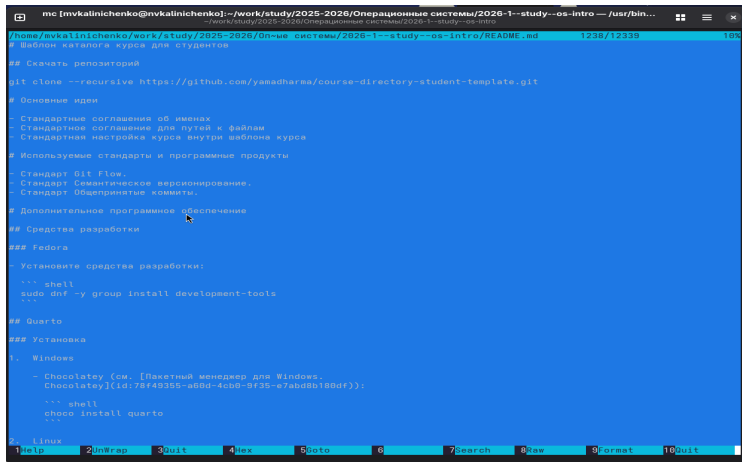


Рисунок 9: Дерево каталогов



The screenshot shows the Midnight Commander (MC) interface. The title bar indicates the current directory is `/usr/bin...`. The main window displays the contents of a file named `README.md` located at `/home/mvkalinichenko/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/README.md`. The file size is 1238/12339 bytes and it is 100% readable. The content of the file is as follows:

```
# Шаблон каталога курса для студентов

## Скачать репозиторий

git clone --recursive https://github.com/yamadharma/course-directory-student-template.git

# Основные идеи

- Стандартные соглашения об именах
- Стандартное соглашение для путей к файлам
- Стандартная настройка курса внутри шаблона курса

# Используемые стандарты и программные продукты

- Стандарт Git Flow.
- Стандарт Семантическое версионирование.
- Стандарт Общепринятые коммиты.

# Дополнительное программное обеспечение

## Средства разработки

### Fedora

- Установите средства разработки:

```shell
sudo dnf -y group install development-tools
```

### Quarto

### Установка

1. Windows

- Chocolatey (см. [Пакетный менеджер для Windows. Chocolatey](id:78f49355-a68d-4cb9-9f35-e7abd8b189df)):

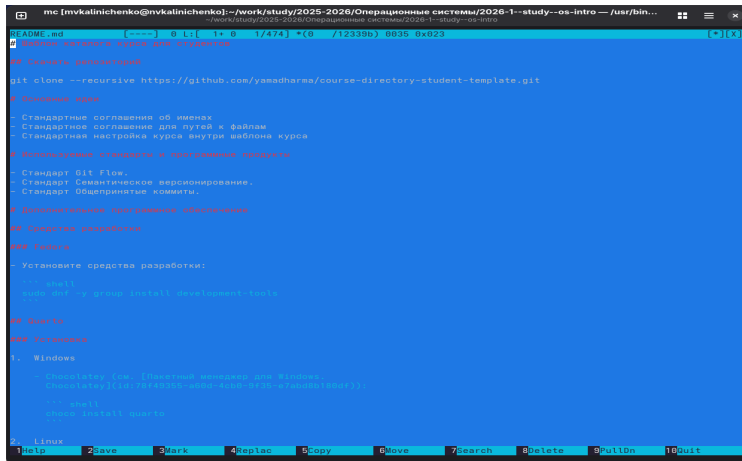
```shell
choco install quarto
```

2. Linux
```

At the bottom of the window, there is a toolbar with icons for various actions: Help, UnWrap, Quit, Hex, Boto, a numeric keypad (0-9), Search, Row, Format, and Quit.

Рисунок 10: Просмотр содержимого текстового файла

Работа с Midnight Commander



```
mc [nvkalinichenko@nvkalinichenko]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro — /usr/bin...
README.md [-----] 0 L: [ 1+ 0 1/474 ] *(0 /12339b) 0035 0x023 [*)(X)]
# Шаблоны каталогов курса для студентов

## Создание репозитория

git clone --recursive https://github.com/yamadharma/course-directory-student-template.git

# Основные идеи

- Стандартные соглашения об именах
- Стандартное соглашение для путей к файлам
- Стандартная настройка курса внутри шаблона курса

# Внедряемые стандарты и программные продукты

- Стандарт Git Flow.
- Стандарт Семантическое версионирование.
- Стандарт Общепринятые коммиты.

# Дополнительное программное обеспечение

## Средства разработки

### Fedora

- Установите средства разработки:

  % shell
  sudo dnf -y group install development-tools

### Quarta

### Windows

1. Windows

   - Chocolatey (см. [Пакетный менеджер для Windows.
     Chocolatey](id:78f49355-a600-4c0b-8f35-e7ab0b6180d1)):

     % shell
     choco install quarta

2. Linux
```

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

Работа с Midnight Commander

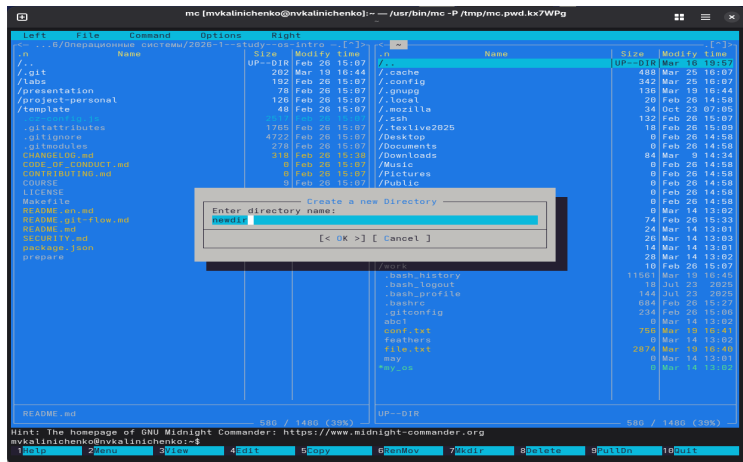


Рисунок 12: Создание каталога

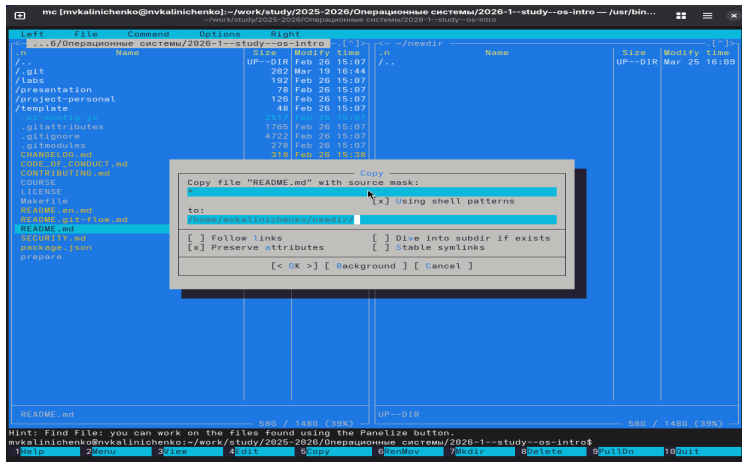


Рисунок 13: Копирование в файлов в созданный каталог

Работа с Midnight Commander

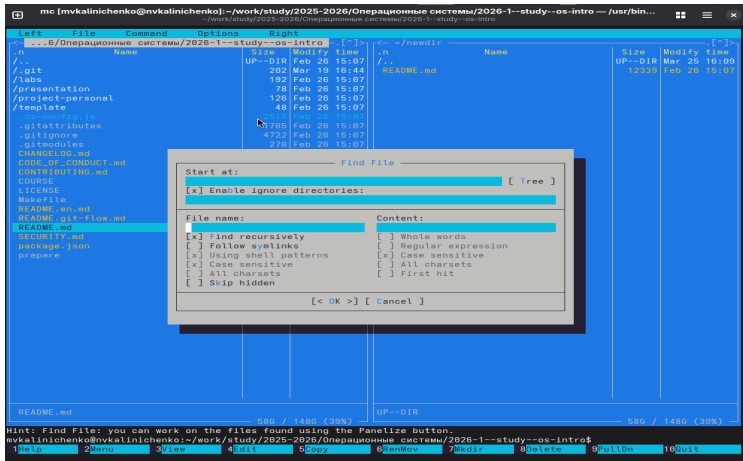


Рисунок 14: Поиск файлов

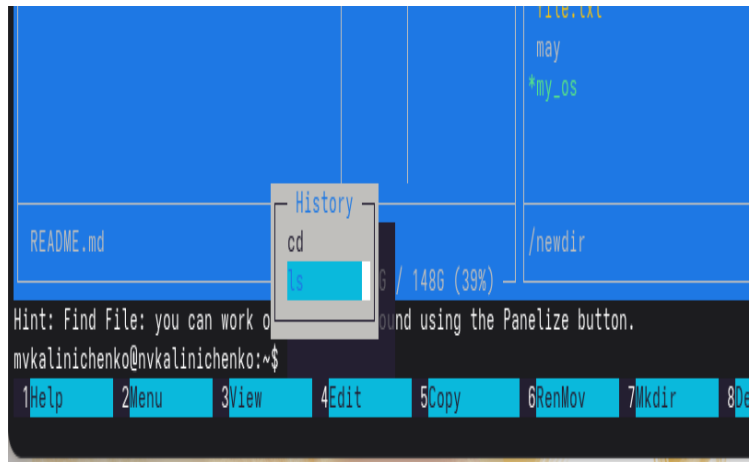


Рисунок 15: История команд

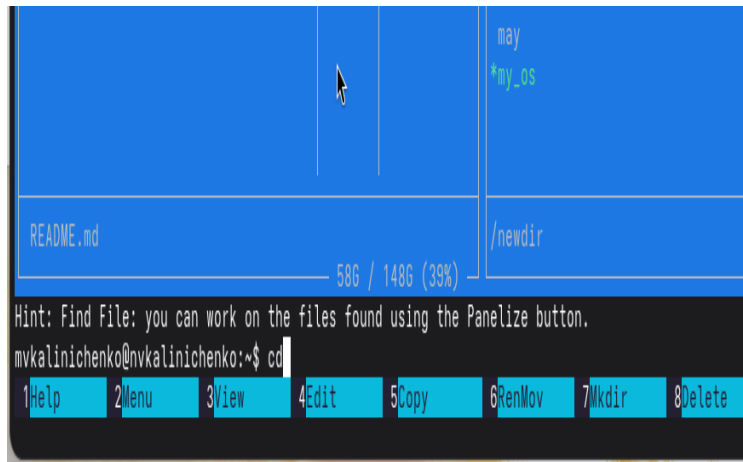
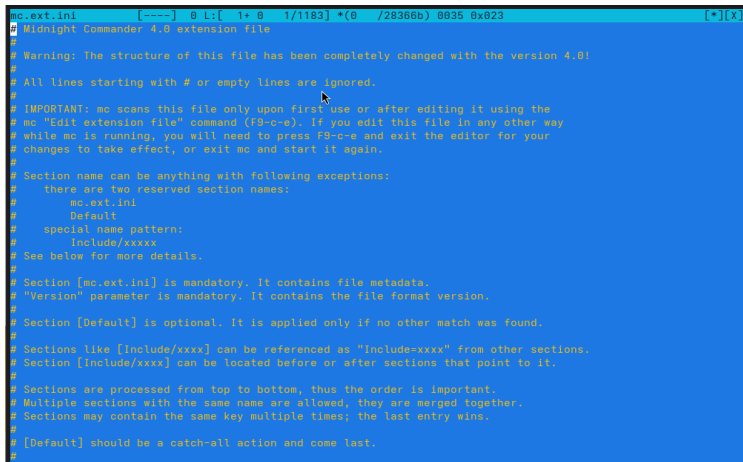
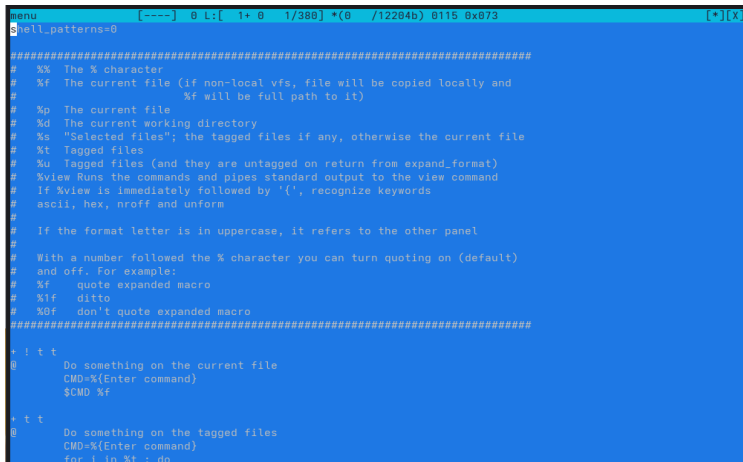


Рисунок 16: Переход в домашний каталог



```
mc.ext.ini      [----]  0 L: [ 1+ 0 1/1183] *(0 /28366b) 0035 0x023      [*][X]
# Midnight Commander 4.0 extension file
#
# Warning: The structure of this file has been completely changed with the version 4.0!
#
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#       mc.ext.ini
#       Default
#   special name pattern:
#       Include/xxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [Include/xxxx] can be referenced as "Include=xxxx" from other sections.
# Section [include/xxxx] can be located before or after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# Multiple sections with the same name are allowed, they are merged together.
# Sections may contain the same key multiple times; the last entry wins.
#
# [Default] should be a catch-all action and come last.
```

Рисунок 17: Просмотр файла расширений

A screenshot of the Midnight Commander file menu. The window title bar shows 'menu' and some system information. The menu content is displayed on a blue background with white text. It starts with 'shell_patterns=0' and a list of macros with their descriptions. At the bottom, there are two menu items: '+ ! t t' and '+ t t', each followed by a description and a command template.

```
menu [----] 0 L:[ 1+ 0 1/380] *(0 /12204b) 0115 0x073 [X]
shell_patterns=0

#####
# %% The % character
# %f The current file (if non-local vfs, file will be copied locally and
#           %f will be full path to it)
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if any, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '{', recognize keywords
#   ascii, hex, proff and unform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
# %f   quote expanded macro
# %if  ditto
# %0f  don't quote expanded macro
#####

+ ! t t
0   Do something on the current file
    CMD=%{Enter command}
    $CMD %f

+ t t
0   Do something on the tagged files
    CMD=%{Enter command}
    for i in %t ; do
```

Рисунок 18: Просмотр файла меню

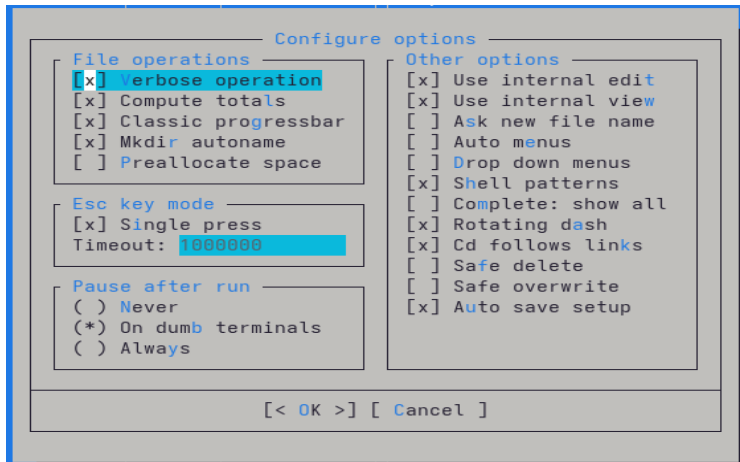


Рисунок 19: Конфигурация

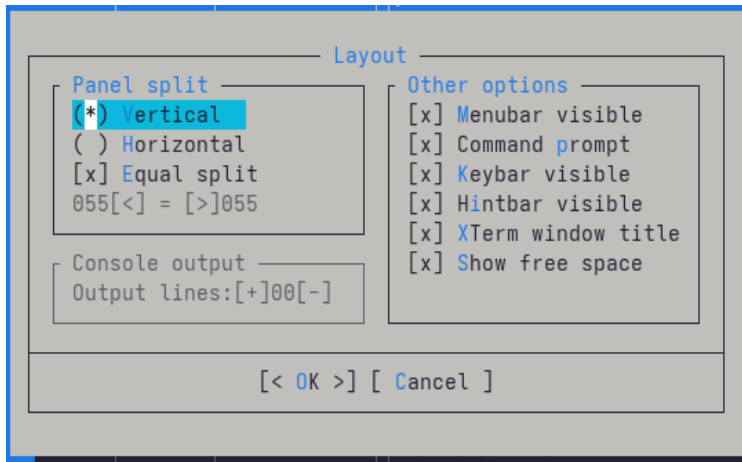


Рисунок 20: Внешний вид

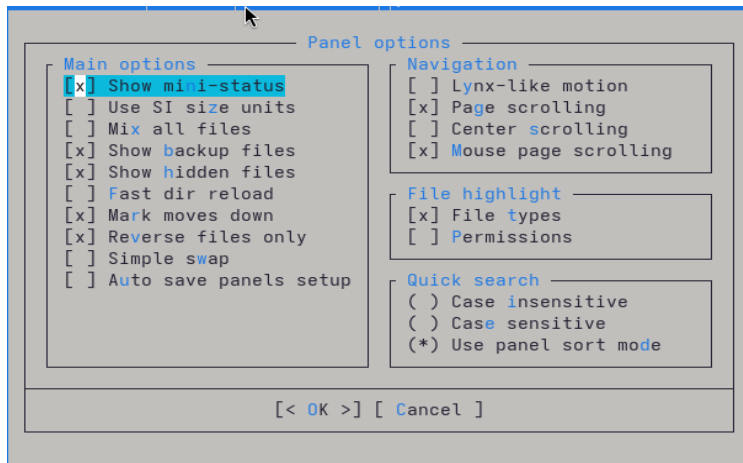


Рисунок 21: Настройки панелей

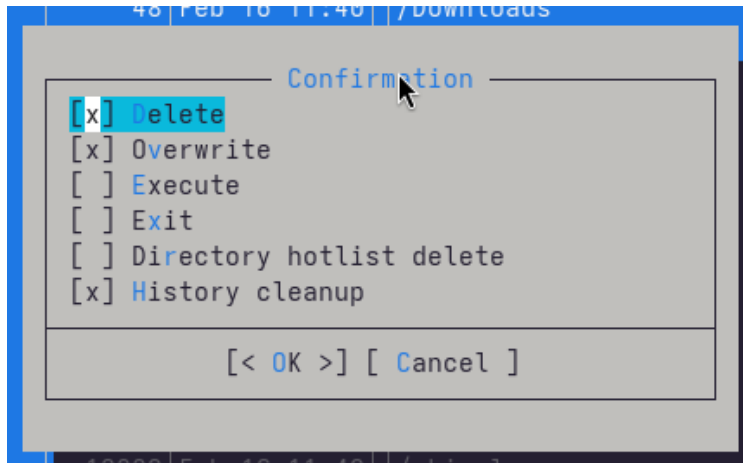


Рисунок 22: Подтверждение

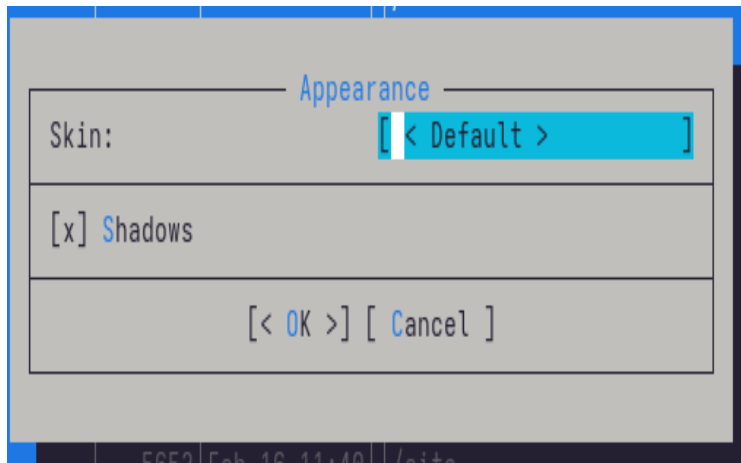


Рисунок 23: Оформление

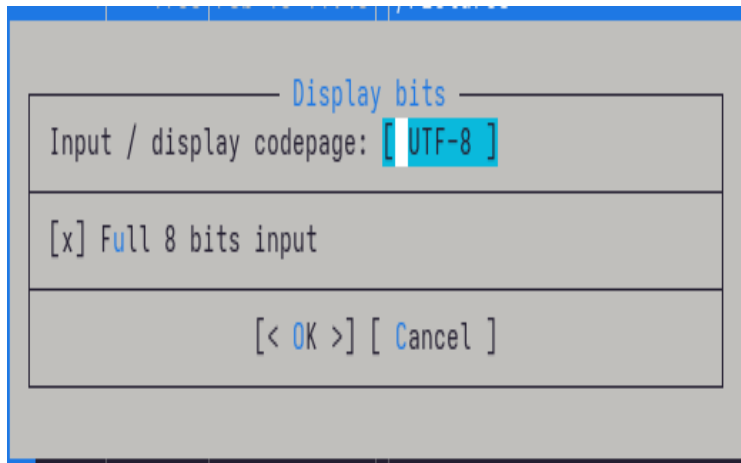


Рисунок 24: Кодировка символов

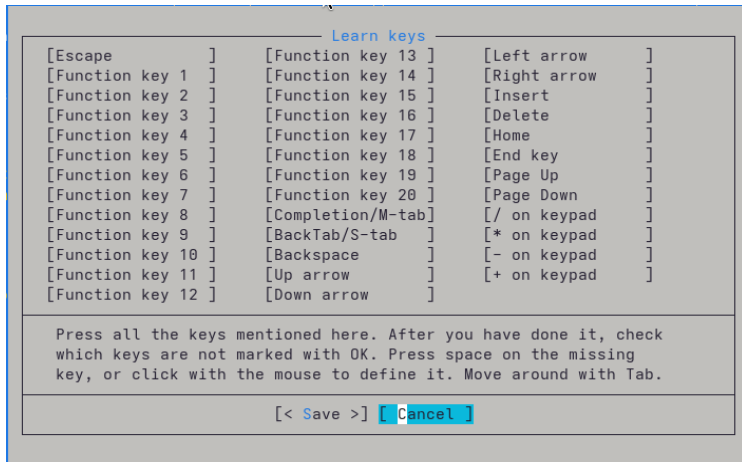
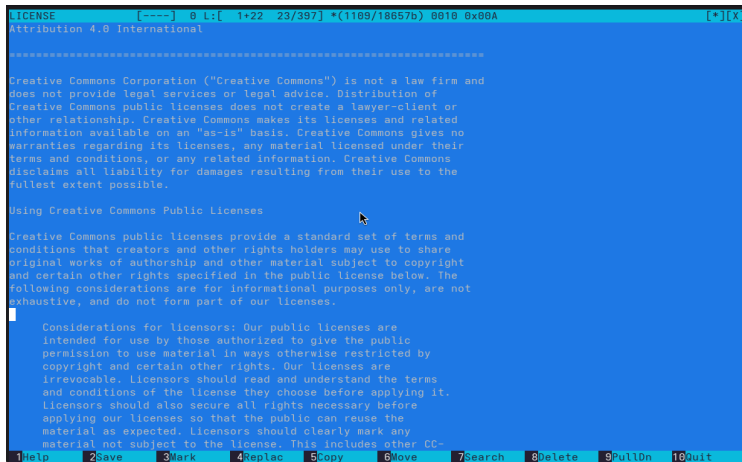


Рисунок 25: Распознавание клавиш



```
LICENSE      [-----]  8 L:[ 1+22 23/397] *(1109/18657b) 0010 0x00A      [*][X]
Attribution 4.0 International

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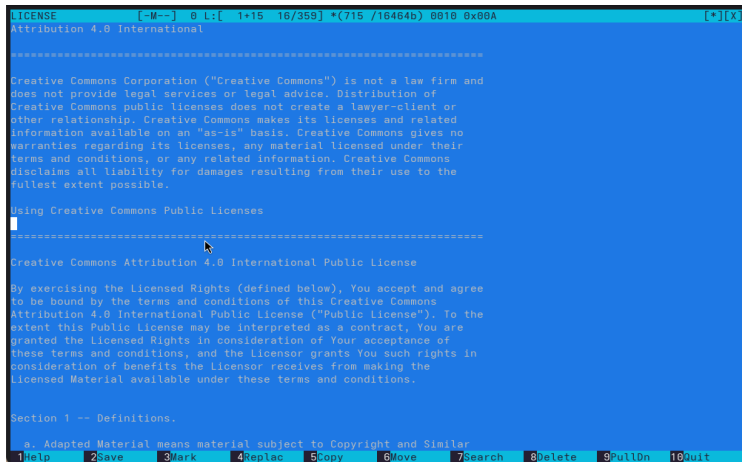
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  material not subject to the license. This includes other CC-
```

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit

Рисунок 26: Файл с текстом

Работа с редактором Midnight Commander



```
LICENSE      [-M--]  8 L:[ 1+15 16/359] *(715 /16464b) 0010 0x00A      [*][x]
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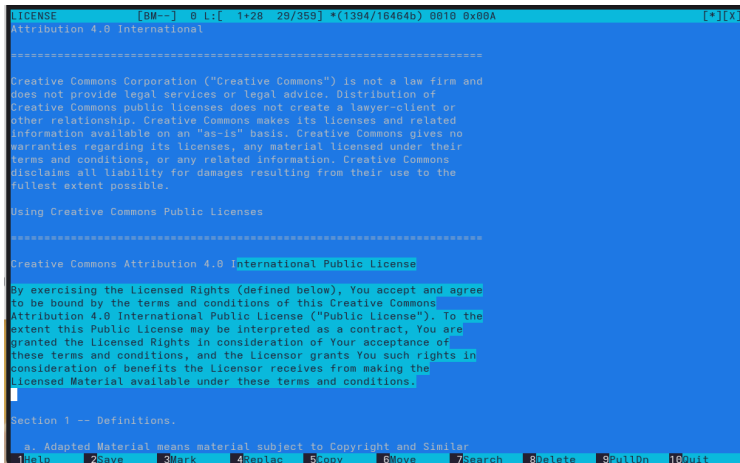
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1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 27: Файл с текстом



```
LICENSE      [BM--]  0 L:[ 1+28 29/359] *(1394/16464b) 0010 0x00A      [*][X]
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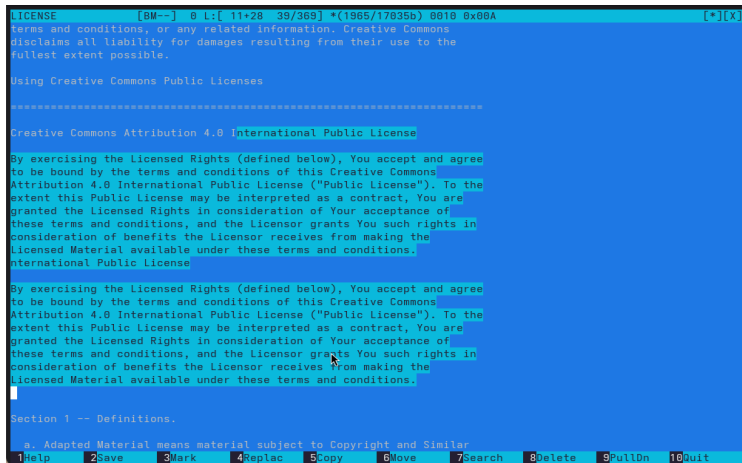
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1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullOn     10Quit
```

Рисунок 28: Копирование фрагмента



The screenshot shows the Midnight Commander file manager and text editor. The top status bar displays 'LICENSE [BM--] 0 L: [11*28 39/369] *(1965/17035b) 0010 0x00A [*][X]'. The main window shows the text of a Creative Commons Attribution 4.0 International Public License. The text is as follows:

```
LICENSE [BM--] 0 L: [11*28 39/369] *(1965/17035b) 0010 0x00A [*][X]
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consideration of benefits the Licensor receives from making the
Licensed Material available under these terms and conditions.

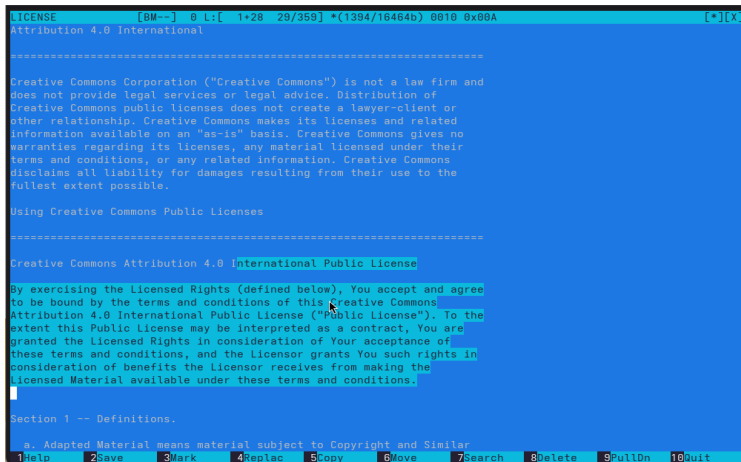
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```

The bottom status bar shows a menu with the following items: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, 10Quit.

Рисунок 29: Сохранение

Работа с редактором Midnight Commander



```
LICENSE      [BM--]  0 L:[ 1+20  29/359] *(1394/16464b) 0010 0x00A      [*][X]
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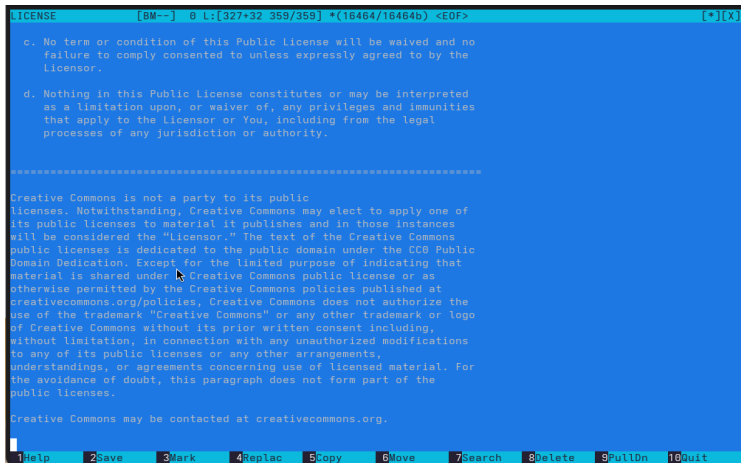
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1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 30: Отмена



```
LICENSE      [BM--]  0 L:[327+32 359/359] *(16464/16464b) <EOF>      [*][X]

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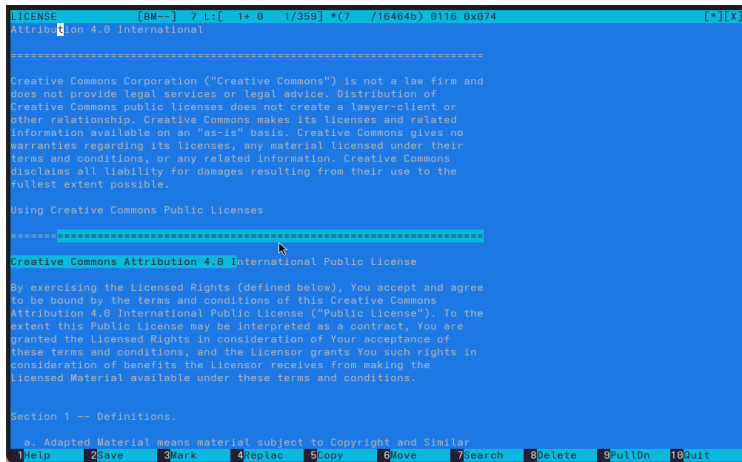
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1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 31: Переход в конец файла



```
LICENSE      [BM--] 7 L:[ 1+ 0 1/359] *(7 /16464b) 0116 0x074      [*][x]
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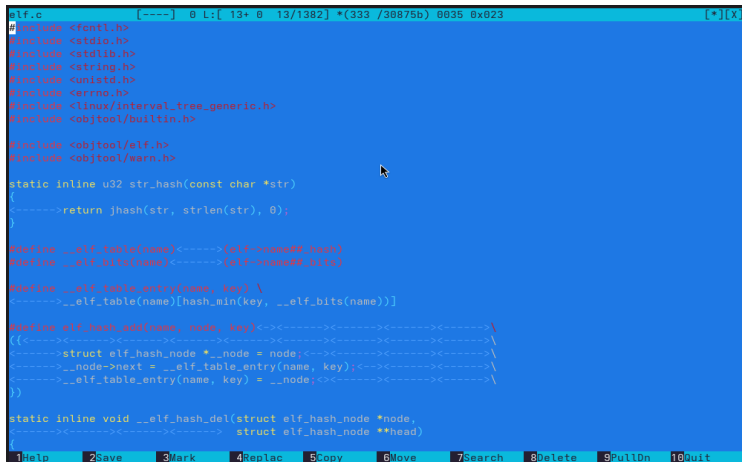
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Section 1 -- Definitions.

a. Adapted Material means material subject to Copyright and Similar

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 32: Переход в начало файла



```
elf.c [----] 0 L: [ 13+ 0 13/1382] *(333 /30875b) 0035 0x023 [*][X]
#include <fcntl.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <errno.h>
#include <linux/interval_tree_generic.h>
#include <objtool/builtin.h>

#include <objtool/elf.h>
#include <objtool/warn.h>

static inline u32 str_hash(const char *str)
{
    ----->return jhash(str, strlen(str), 0);
}

#define __elf_table(name)----->{elf->name##_hash}
#define __elf_bits(name)----->{elf->name##_bits}

#define __elf_table_entry(name, key) \
    ----->__elf_table(name)[hash_min(key, __elf_bits(name))]

#define elf_hash_add(name, node, key)----->----->----->----->\
    ({----->----->----->----->----->----->----->----->\
    ----->struct elf_hash_node *__node = node;----->----->----->----->\
    ----->__node->next = __elf_table_entry(name, key);----->----->----->----->\
    ----->__elf_table_entry(name, key) = __node;----->----->----->----->\
    })

static inline void __elf_hash_del(struct elf_hash_node *node,
    ----->----->----->----->-----> struct elf_hash_node **head)
{
}

1Help 2Save 3/ark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 33: Файл с программой

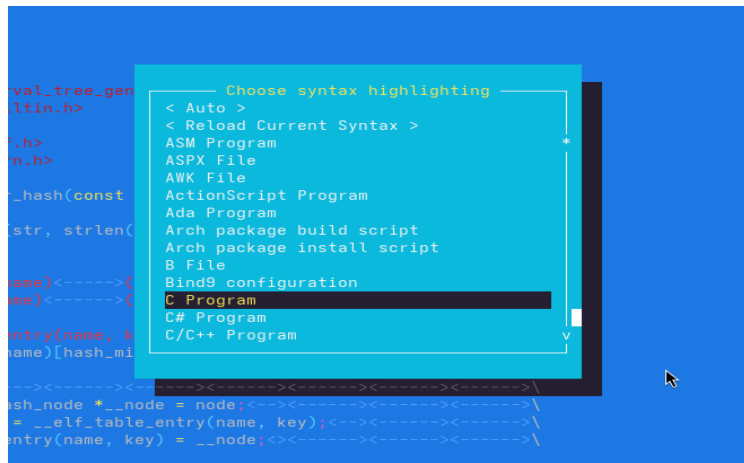


Рисунок 34: Цветовыделение синтаксиса

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.